

The Trust Layer

An agentic trust check for small-data biology models

Point it at a dataset and analysis notebook. It audits for data leakage, re-runs an honest cross-validation, and tells a biologist what their model is really worth.

Data-leakage audit

Leakage-safe CV

Calibrated LogReg

Plain-language report

The quiet reproducibility problem

A bench scientist builds a classifier, sees 0.95 AUC — and it never replicates.

Small n, huge p

Datasets are often $n < 200$ with thousands of features. Selecting the 'best' features on the full data manufactures signal from noise.

Subject leakage

Repeated measures from one patient straddle the train/test split. The model memorises the subject, not the biology.

Batch confounding

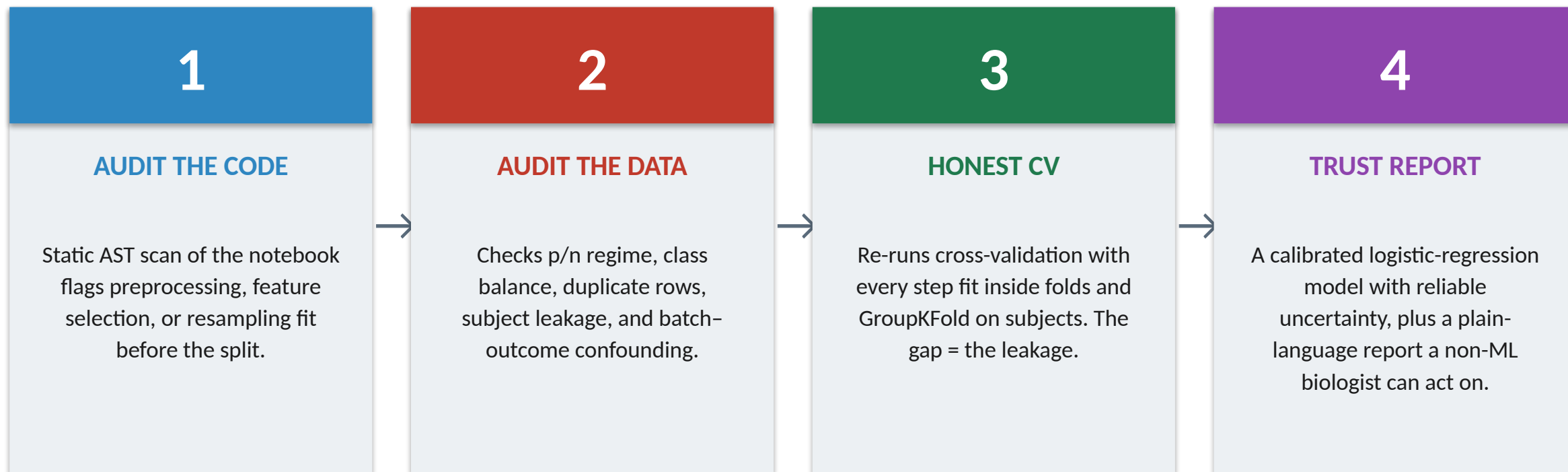
A plate or site variable correlates with the outcome. The model reads batch signatures instead of disease.

Leaky preprocessing

Scaling or feature selection fit on the whole dataset lets test rows inform the pipeline — before any prediction is made.

What we built: an automated trust layer

One call runs the whole pipeline — the scientist never has to self-declare their mistakes.



Design principle: naive performance – honest performance = the leakage.

The whole thing is one function call

A scientist points it at their data and their notebook — no ML expertise required.

```
result = trust_audit(  
    X, y,                # samples x features,  
    labels               #  
    groups = subject_ids, # -> subject-safe CV  
    batch = plate_labels, # -> batch-confound  
    check               #  
    notebook = "analysis.ipynb", # -> static code  
    audit               #  
)  
  
print(result["report_md"])
```

What comes back

Code audit

leaks found in the notebook, by line

Data audit

taxonomy findings, critical / warning

CV ladder

naive → honest, step by step

Honest model

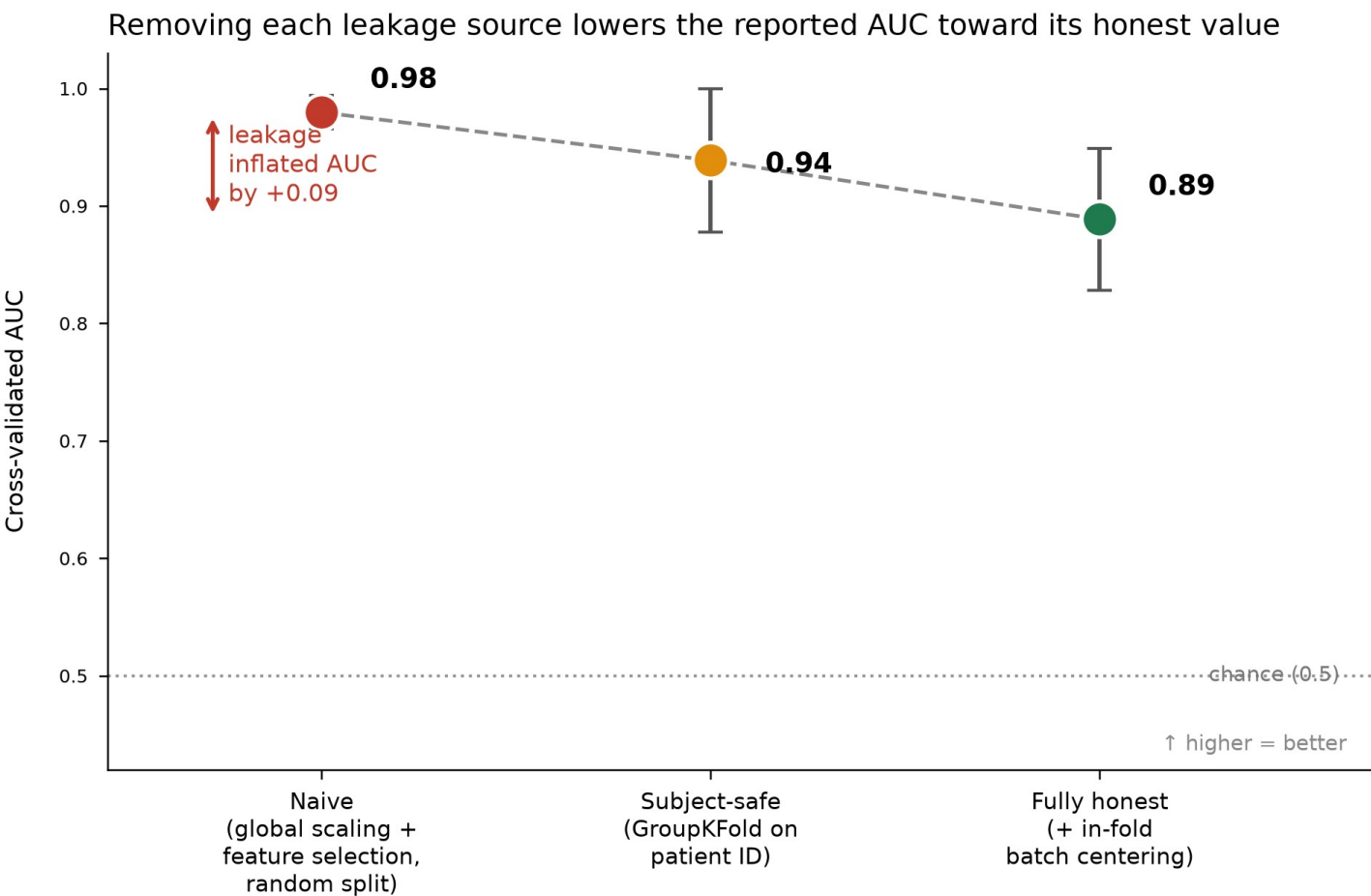
AUC + calibrated Brier score

Plain report

what to trust, what to fix

Result 1 — we catch the inflation and quantify it

Synthetic benchmark with known ground truth: 160 samples, 800 features, 40 subjects.



The ladder

Naive pipeline 0.98

feature selection + scaling on full data

+ subject-safe CV 0.94

GroupKFold keeps patients intact

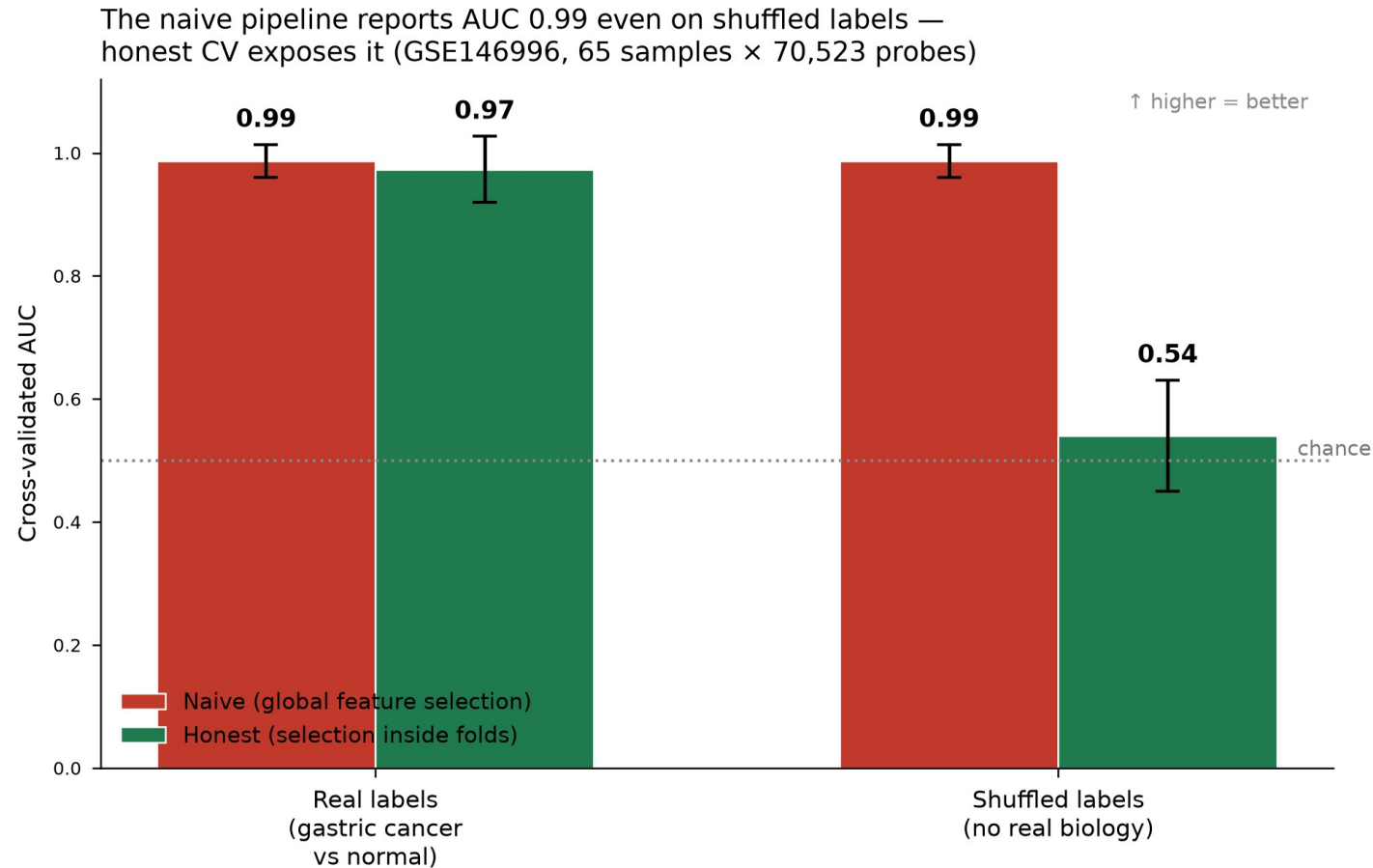
+ batch correction 0.89

fully honest, in-fold centering

Leakage inflated AUC by +0.09

Result 2 — proof on real biology (GEO GSE146996)

Gastric cancer vs normal, 65 samples × 70,523 probes — a real $p \gg n$ dataset.



The killer demo

Shuffle the labels so NO biology remains. The naive pipeline still reports AUC 0.99 — on pure noise.

Naive, shuffled labels

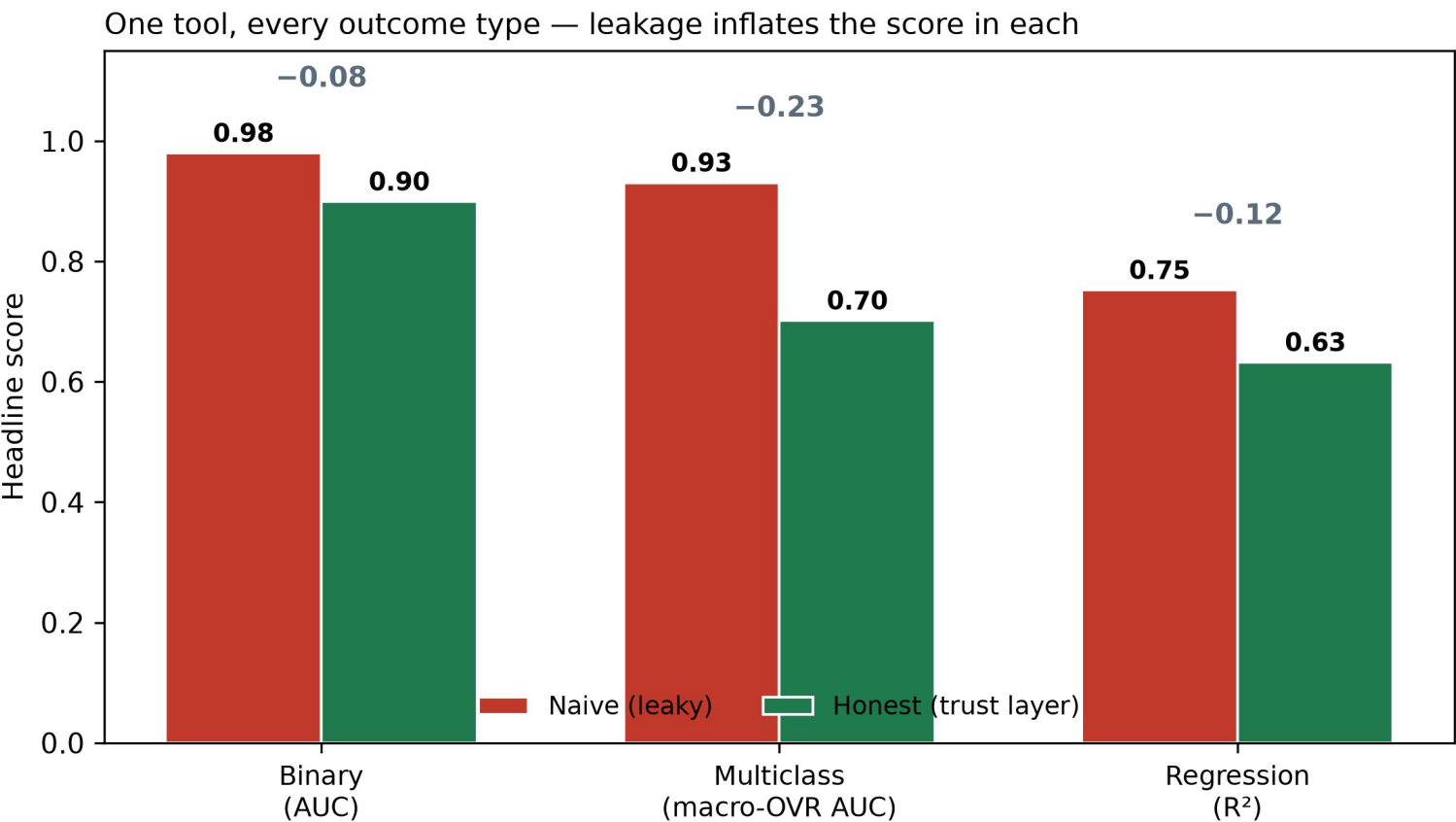
AUC 0.99 → pure artifact

Honest CV, shuffled labels

AUC 0.54 → chance (correct)

Result 3 — one tool, every outcome type

Auto-detects the task from the labels. Same leakage story holds throughout.



Coverage

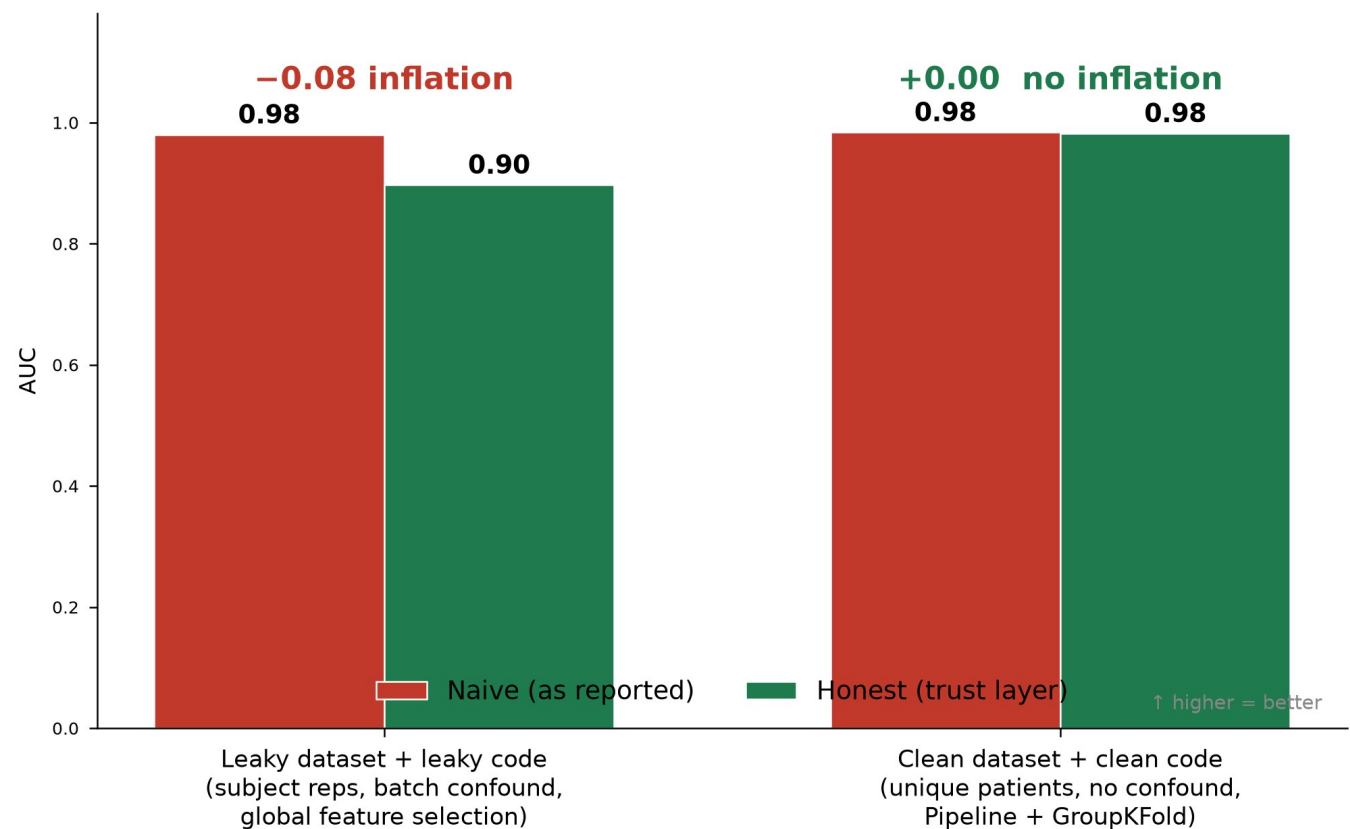
- Binary**
AUC 0.98 → 0.90 (naive → honest)
Calibrated LogReg
- Multiclass**
AUC 0.93 → 0.70 (naive → honest)
macro-OVR, balanced acc.
- Regression**
R² 0.75 → 0.63 (naive → honest)
Ridge, RMSE / MAE

+ nested CV for tuning honesty

Result 4 — it doesn't cry wolf

The real test of a trust tool: does it stay quiet when the analysis is actually sound?

Real leakage is flagged — a clean analysis correctly shows none



Leaky data + leaky code

4 critical findings
naive 0.98 → honest 0.90
Inflation flagged: +0.08

Clean data + clean code

0 critical findings
naive 0.98 → honest 0.98
No inflation: +0.00

Why it matters



Catches what humans miss

The two hardest leaks to self-report — preprocessing and feature selection before the split — are found automatically by reading the code.



Honest, not alarmist

On real signal it reports near-zero inflation. It distinguishes genuine biology from artifact instead of crying wolf.



Built for the bench

A non-ML biologist gets a plain-language verdict: here is your real performance, here is what inflated it, here is how confident the model actually is.



Shipped as a reusable skill

Binary / multiclass / regression + nested CV, one call, validated on real GEO data — ready to run on the next dataset.

The Trust Layer — because a 0.95 AUC should mean something.

Appendix

Detailed data analysis — leakage taxonomy, calibration, and methods

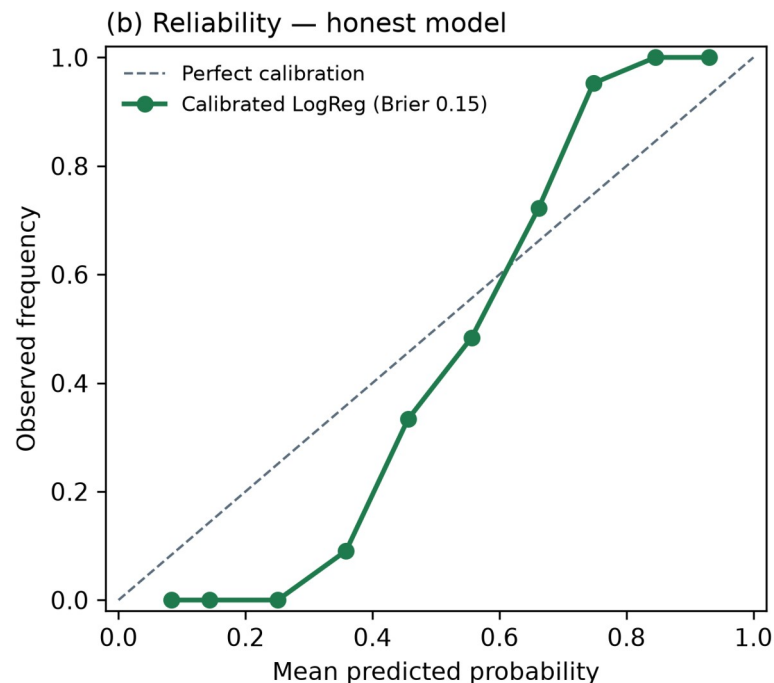
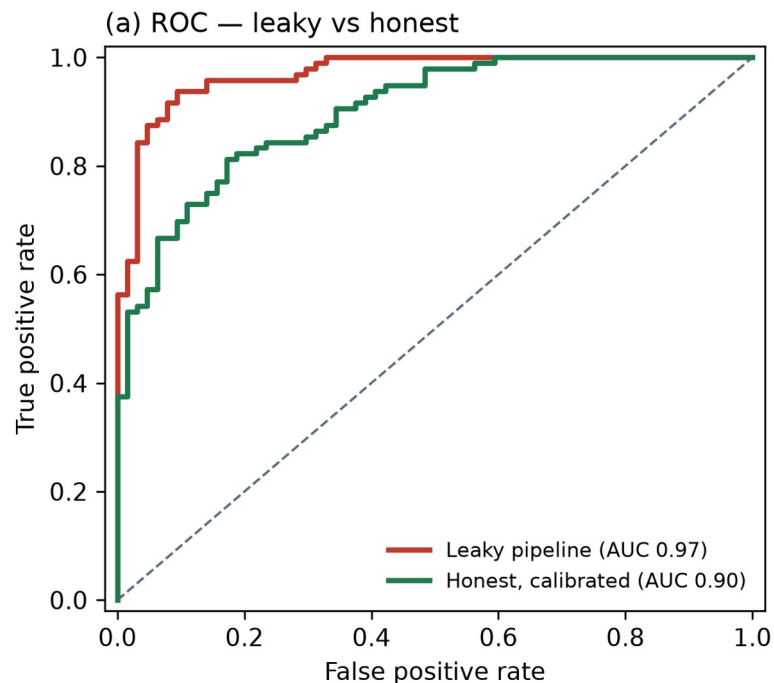
Appendix A — leakage audit on the benchmark

Synthetic dataset: 160 samples, 800 features, 40 subjects × 4 replicates. 4 critical, 1 warning.

Check	Severity	Finding
subject_leakage	CRITICAL	40 subjects, 160 rows (4 reps each). A random split puts the same subject on both sides.
batch_confound	CRITICAL	Batch confounded with outcome (Cramér's V = 0.67, p = 1.4e-17).
leaky_feature_selection	CRITICAL	Selection on full data before split — with $p \gg n$ manufactures 0.9+ AUC from noise.
leaky_preprocessing	CRITICAL	Scaling fit on full dataset; test rows informed the transform.
dimensionality	WARNING	800 features vs 160 samples ($p/n = 5.0$). Nested CV mandatory.
class_balance	OK	Minority class 40% — reasonably balanced.
duplicate_rows	OK	No exact duplicate feature rows.

Appendix B — honest performance & calibration

Calibrated logistic regression, leakage-safe out-of-fold predictions on the benchmark.



Calibrated LogReg (honest)

Honest AUC **0.899**

Leaky AUC **0.970**

Brier (honest) **0.146**

Brier (leaky) **0.072**

Log-loss (honest) **0.456**

Probabilities calibrated inside each fold — Brier-reliable, leakage-safe.

Appendix C — method & reproducibility

Every number in this deck is reproducible from saved artifacts.

Honest cross-validation

- Scaling, feature selection, and per-batch centering fit INSIDE each training fold only
- GroupKFold on subject ID — no subject spans train and test
- Naive pipeline fits the same steps on the full data (the deliberate leak)

Model & calibration

- Calibrated logistic regression for binary; Ridge / multinomial LogReg otherwise
- Out-of-fold probabilities give unbiased AUC + Brier calibration
- Nested CV measures the optimism of hyperparameter selection

Notebook static analysis

- AST parse of .ipynb / .py — no code executed
- Flags fits before the split; recognises Pipeline + GroupKFold as safe

Validation data

- Synthetic benchmark with known ground truth (25 true features)
- Real: GEO GSE146996 — gastric cancer vs normal, 65 × 70,523